

ASEN 5264 Decision Making under Uncertainty
Exam 1 (2026): Probabilistic Models and MDPs

August 24, 2026

SOLUTION

Question 1

Solution:

(a) Yes. $C \leftarrow B \rightarrow E$ is a fork and B is in the evidence set. Therefore this path (the only path) is d-separated by $\{B\}$. Since all paths are d-separated, we can conclude that the conditional independence holds.

(b) No. $A \rightarrow C \leftarrow B \rightarrow E$ contains two triples:

- $A \rightarrow C \leftarrow B$ is a collider, and evidence C makes this triple active
- $C \leftarrow B \rightarrow E$ is a fork, but B is not in the evidence, so this triple is also active

This is an active path from A to E . Thus the structure does not imply $A \perp E \mid C$.

(c)

Note that E is conditionally independent from all other nodes given B , so it is not relevant.

$$\begin{aligned} P(A = 1 \mid D = 1) &= \frac{P(A = 1, D = 1)}{P(D = 1)} && \text{(Definition of conditional probability)} \\ &= \frac{\sum_{b,c} P(A = 1, B = b, C = c, D = 1)}{\sum_{a,b,c} P(A = a, B = b, C = c, D = 1)} && \text{(Marginalization)} \\ &= \frac{\sum_{b,c} P(D = 1 \mid C = c) P(C = c \mid A = 1, B = b) P(B = b) P(A = 1)}{\sum_{a,b,c} P(D = 1 \mid C = c) P(C = c \mid A = a, B = b) P(B = b) P(A = a)} && \text{(Bayes net chain rule)} \end{aligned}$$

Question 2

Solution:

(a)

$$\mathcal{S} = \{F, L\}, \quad \mathcal{A} = \{E, D\}$$

where F = fall and L = landed (stoked), E = easy trick and D = difficult trick.

$$R(s, a, s') = \begin{cases} 70 & \text{if } a = E \text{ \& } s' = L \\ 90 & \text{if } a = D \text{ \& } s' = L \\ 0 & \text{otherwise} \end{cases} \quad \gamma = 0.9$$

The transition matrices (rows correspond to current state and columns to next state):

$$T^E = \begin{bmatrix} 0.2 & 0.8 \\ 0.2 & 0.8 \end{bmatrix}, \quad T^D = \begin{bmatrix} 0.3 & 0.7 \\ 0.2 & 0.8 \end{bmatrix}$$

(b) The expected reward at state F is $R(F, E) = 0.8 \cdot 70 = 56$ and $R(F, D) = 0.7 \cdot 90 = 63$, so the greedy action is D , attempting the difficult trick. The snowboarder might want to take action E from this state because it has higher probability of getting her to the landed (stoked) state from which she can potentially get more reward in the future.

(c)

$$\tilde{\pi}(s) = \begin{cases} E & \text{if } s = F \\ D & \text{if } s = L \end{cases}$$

(d) Let

$$T^{\tilde{\pi}} = \begin{bmatrix} 0.2 & 0.8 \\ 0.2 & 0.8 \end{bmatrix}, \quad R^{\tilde{\pi}} = \begin{bmatrix} 70 \cdot 0.8 \\ 90 \cdot 0.8 \end{bmatrix}, \quad R^D = \begin{bmatrix} 90 \cdot 0.7 \\ 90 \cdot 0.8 \end{bmatrix}$$

First calculate the vectors

$$U^{\tilde{\pi}} = (I - \gamma T^{\tilde{\pi}})^{-1} R^{\tilde{\pi}}$$

and

$$U^D = (I - \gamma T^D)^{-1} R^D$$

Then compare the value functions element-wise to determine which policy is better at each state. Since the rider starts in the unstoked (F) state, then it is reasonable to conclude that the policy with the higher first element in its utility vector is better. (Some other definitions of better were also accepted)

(e) **No.** There always exists a deterministic optimal policy for an MDP. Therefore, if this new stochastic $\tilde{\pi}'$ is indeed optimal, there exists at least one other deterministic policy that yields the same optimal value.

Question 3

Solution: TD update: $Q(s, a) \leftarrow Q(s, a) + \alpha (Q_{\text{target}}(s, a) - Q(s, a))$.

(a)

i) **Q-learning:** $Q_{\text{target}}(s_1, a) = r_1 + \gamma \max_{a'} Q(s_2, a') = 0 + 0.9 \cdot \max\{4, 6\} = 5.4$.

$$Q_{\text{new}}(1, a) = 1.0 + 0.5(5.4 - 1.0) = 1.0 + 2.2 = 3.2.$$

ii) **SARSA:** next action in s_2 is a , so $Q_{\text{target}}(s_1, a) = r_1 + \gamma Q(s_2, a) = 0 + 0.9 \cdot 4 = 3.6$.

$$Q_{\text{new}}(1, a) = 1.0 + 0.5(3.6 - 1.0) = 1.0 + 1.3 = 2.3.$$

For both methods, $Q_{\text{target}}(s_2, a) = r_2 + \gamma Q(3, \cdot) = 5 + 0.9 \cdot 0 = 5$, so

$$Q_{\text{new}}(2, a) = 4.0 + 0.5(5 - 4.0) = 4.0 + 0.5 = 4.5,$$

for both methods as well.

(b)

The reward is approximated with

$$\hat{R}(s, a) = \frac{\rho(s, a)}{N(s, a)}$$

where ρ is the total reward received over all visits to (s, a) in all trajectories, and $N(s, a)$ is the number of visits in all trajectories. 0 is used if the state is terminal.

Transitions probabilities are approximated with

$$\hat{T}(s \mid s, a) = \frac{N(s, a, s')}{N(s, a)}$$

where $N(s, a, s')$ is the number of transitions from s to s' after action a in all trajectories.

$$\hat{R}(1, a) = \frac{0 + (-4)}{2} = -2, \quad \hat{R}(2, a) = 5/1 = 5, \quad \hat{R}(3, a) = 0$$

$$\hat{R}(1, b) = 1/1 = 1, \quad \hat{R}(2, b) = -8/1 = -8, \quad \hat{R}(3, b) = 0.$$

$$\hat{T}^a = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\hat{T}^b = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 4

Solution:

(a)

$$U_{\max} = \frac{r_{\max}}{1 - \gamma} = \frac{10}{1 - 0.9} = 100 < 200$$

The maximum achievable value is upper bounded by 100, therefore a value of 200 is *not* achievable.

(b)

Since the max norm (denoted simply with $\|\cdot\|$ in this derivation) is a metric, the triangle inequality yields

$$\|U^* - U_k\| \leq \|U^* - B[U_k]\| + \|B[U_k] - U_k\| \quad (1)$$

$$= \|B[U^*] - B[U_k]\| + \|B[U_k] - U_k\| \quad (\text{since } U^* \text{ is a fixed point, } B[U^*] = U^*) \quad (2)$$

$$\leq \gamma \|U^* - U_k\| + \|B[U_k] - U_k\| \quad (\text{since } B \text{ is a } \gamma \text{ contraction}). \quad (3)$$

Bringing matching terms to the left side,

$$(1 - \gamma) \|U^* - U_k\| \leq \|B[U_k] - U_k\|$$

and

$$\|U^* - U_k\| \leq \frac{\|B[U_k] - U_k\|}{1 - \gamma}.$$

Thus $\|U_k - U^*\|_{\infty} \leq 2$. 3 falls outside of this upper bound, therefore **it is impossible for** $\|U_k - U^*\|_{\infty} = 3$.